



PREDICTIVE MODELING USING MACHINE LEARNING

Heart Disease Prediction using UCI ML Repository Dataset



OBJECTIVE

- Develop a machine learning model to predict heart disease.
- Analyze patient health attributes.
- Compare multiple classification algorithms.

KEY FEATURES

- Data Cleaning & Preprocessing
- Logistic Regression, Decision Tree, Random Forest
- Model Evaluation using Accuracy, Confusion Matrix, and ROC Curve
- Binary Classification (Disease / No Disease)

EXPECTED OUTCOME

- Early detection of heart disease risk
- Comparison of machine learning algorithms
- Selection of the best-performing model
- Support data-driven healthcare decisions

WORKFLOW



DATASET OVERVIEW

Metric	Value
Dataset	Heart Disease (UCI)
Records	303
Features	13
Target Variable	Heart Disease
Classification Type	Binary

ALGORITHMS USED & PERFORMANCE

Algorithm	Accuracy	Performance
Logistic Regression	0.8852459016393442	88.52%
Decision Tree	0.7540983606557377	75.41%
Random Forest	0.8688524590163934	86.89%

BEST PERFORMING MODEL



LOGISTIC REGRESSION
88.52% Accuracy

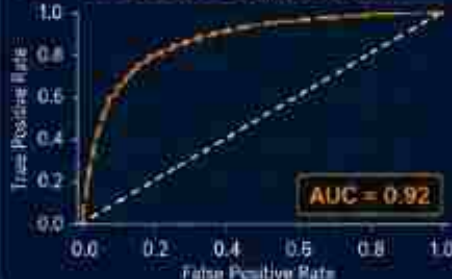
PREDICTION SUMMARY

Best Model	Logistic Regression
Accuracy	88.52%
Dataset Type	Supervised Learning
Prediction Task	Heart Disease Classification

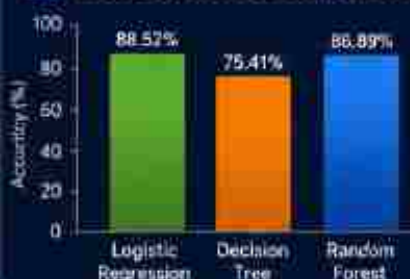
CONFUSION MATRIX (Logistic Regression)

	No Disease (0)	Disease (1)
Actual No Disease (0)	25	4
Actual Disease (1)	3	29

ROC CURVE (Logistic Regression)



MODEL ACCURACY COMPARISON



FINAL OUTCOME

Model Successfully Trained
Best Algorithm: Logistic Regression
Achieved Accuracy: 88.52%
Suitable for Heart Disease Risk Prediction

TECHNOLOGIES USED



Python



Pandas



NumPy



Scikit-learn



Matplotlib



Seaborn